

Nonintrusive Perceptual Audio Quality Assessment for User-Generated Content Using Deep Learning

Deebha Mumtaz , Vinit Jakhetiya , Karan Nathwani, Badri Narayan Subudhi ,
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Abstract—With the boom of social media communication, teleconferencing, and online classes, audiovisual communication over bandwidth strained networks has become an integral part of our lives. Consequently, the growing demand for the quality of experience necessitates developing algorithms to measure and enrich user experience. Prior studies have mainly focused on assessing speech quality and intelligibility with reference to audio quality assessment, while other categories in user-generated multimedia (UGM) are less explored. Moreover, frequency-domain properties of speech and UGM audio are significantly different from each other. Furthermore, there is a lack of a standard dataset for the quality assessment of UGM. Considering these limitations, in this article, we first develop the IIT-JMU-UGM audio dataset consisting of 1150 audio clips, with diverse context, content, and types of degradation commonly observed in real-world scenarios and annotated with the subjective quality scores. Finally, we propose a non-intrusive audio quality assessment metric using a stacked gated-recurrent-unit-based deep learning framework. The proposed model outperforms several baseline methods, including state-of-the-art non-intrusive and intrusive approaches. The resulting Pearson's correlation coefficient of 0.834 indicates that the proposed method efficiently mirrors human auditory perception.

Index Terms—Audio quality assessment, deep learning, gated recurrent unit (GRU), non-intrusive quality metric, user-generated multimedia (UGM).

I. INTRODUCTION

IN RECENT years, there has been an exponential growth in digital technology, resulting in rapid creation and

transmission of user-generated multimedia (UGM) content. With the onset of Coronavirus disease 2019, the world has further embraced multimedia technology for online learning, work-from-home tasks, video conferencing, entertainment, communication, information dissemination, health-related services, online shopping, and many other applications. Millions of hours of videos are daily shared on social media platforms, video conferencing applications, and other video content streaming websites.

In the multimedia domain, extensive research has been done on image quality assessment [1]–[3]. However, only a few algorithms have been proposed for audio quality assessment. Audio quality metrics are divided into two types: intrusive and non-intrusive. Intrusive metrics require both degraded and reference signals to derive perceptually weighted distance to estimate the quality score. Perceptual evaluation of speech quality (PESQ) [4] is an intrusive approach for the quality assessment of narrowband speech signals. Perceptual objective listening quality assessment [5] determines the perceptual quality for narrowband as well as super-wideband signals. Hines *et al.* [6] proposed an intrusive speech metric based on the spectrotemporal measure of similarity between the given samples. Later, Taal *et al.* [7] proposed an intrusive technique based on the intermediate intelligibility metric using discrete-Fourier-transform-based time–frequency decomposition. Roux *et al.* [8] proposed a scale-invariant signal-to-distortion ratio for quality assessment of speech enhancement and source separation algorithms. More recently, Mittag *et al.* [9] have proposed a model based on the convolution neural network (CNN) and the self-attention network for speech quality assessment. Chinen *et al.* [10] proposed an intrusive model using gammatone spectrogram for determining audio quality. The main limitation of all these intrusive algorithms is the requirement of a reference signal, which may not be accessible in real-world scenarios. Conversely, non-intrusive metrics do not require the reference signal and, hence, mimic realistic situations better. To this end, Falk *et al.* [11] evaluated the quality of dereverberated and reverberant speech using spectral modulation. Furthermore, the metric in [12] computes the energy histogram to determine noise elements in a given signal. In [13], a technique is proposed to estimate the signal-to-noise ratio of speech signals by evaluating their amplitude distribution.

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TABLE I
DESCRIPTION OF THE EXISTING AUDIO DATASETS

Dataset	Purpose	Type of Distortions	Year	Availability	Sample Size	Subjective Test
TIMIT [14]	Speech recognition applications	None	1993	Available under Licence	5.4 hours	-
ITU-T P Suppl. 23 [15]	Objective voice quality assessment	Narrowband speech degradation, environmental noise, audio encoding, and channel degradation	1998	Available under Licence	-	Yes
SPINE [16]	Speech recognition in military setup.	Noisy military environment	2000	Available under Licence	12 hours	-
NOIZUS [17]	Assessment of speech enhancement techniques	Various background noises	2006	Available	960 samples	Yes
EBU-SQAM [18]	Sound quality assessment	-	2008	Available	70 samples	-
Creusere's [19]	Audio quality assessment	Changing bit rates	2008	-	48 samples	Yes
Live Music dataset [20]	Live music recording quality assessment	Amplitude compression, amplification, band pass filtering, white noise, and crowd noise	2013	-	2900 samples	Yes
ACE Challenge [21]	Estimation of acoustic parameters	Reverberations, multi-channel noise, different signal to noise ratio	2015	Available	4500 samples	-
CoreSV14 [22]	Evaluation of various codec	Distortions due to different types of Codec used	2014	Available	40 samples	Yes
REVERB Challenge [23]	Evaluation of automatic speech recognition and enhancement techniques	Reverberant room responses and environmental noise	2016	Available	-	Yes
Fazenda's [24]	Audio quality assessment	Background noise	2016	Not Openly Available	128 samples	Yes
Avila's [25]	Speech quality assessment	Room impulse response, background noise	2019	-	10,000 samples	Yes
IIT-JMU-UGM Audio Dataset (Proposed)	User generated multimedia audio quality assessment	Various background noises, different bit rates	2021	Will be open source	1150 samples	Yes

“—” indicates unavailability of relevant information.

Lately, deep neural network (DNN)-based techniques have paved the way in the domain of quality analysis of audio signals. Lo *et al.* [26] proposed a convolution and recurrent neural network (RNN)-based model for assessing voice conversion. In [27], a CNN was used to determine the utterance-level and system-level quality score for synthetic speech. Schwind *et al.* [28] evaluated the effect of stalling in music streaming on the quality of experience. The main disadvantage of these methods is that they generally do not exhibit high performance compared to intrusive techniques. Lately, the transform learning technique [29], [30] has been used in a few signal processing applications, which have shown superior performance in multiple domains. Future work could explore the use of these methods for real-time audio quality assessment.

Quality estimation of audio extracted from UGM data enveloping domains, such as music, casual conversations, natural sounds, and home videos, has been less explored. One of the primary reasons behind this is the challenge of creating a relevant dataset. To address this limitation, in this article, we first propose a UGM dataset with diverse content. Second, we propose a novel non-intrusive audio quality estimation algorithm building on the gated recurrent unit (GRU) architecture. The main contributions of this article are as follows.

- 1) We analyze the existing databases and identify the need for a new comprehensive quality assessment database for the UGM.
- 2) We design and develop a novel and comprehensive large-scale database called IIT-JMU-UGM audio dataset, consisting of 1150 audio clips extracted from UGM clips representing diverse context, content, and variety of

distortions. The database is annotated for subjective quality scores. The proposed database will be open source to the research community.¹

- 3) We design and develop a non-intrusive stacked GRU model with multiple audio features as input for audio quality assessment metric that correlates well with the human perceptual auditory system outperforming the existing state-of-the-art methods achieving Pearson's linear correlation coefficient (PLCC) of 0.834.

The rest of this article is organized as follows. Section II describes the existing audio datasets and the need for developing a new UGM audio dataset. Section III describes the development of the proposed IIT-JMU-UGM audio dataset. The proposed method for UGM audio quality assessment is explained in Section IV and results are discussed in Section V. Finally, Section VI concludes this article.

II. EXISTING AUDIO DATABASES

In this section, we introduce some of the well-known audio and speech datasets that have been employed for various applications. Table I summarizes the key details of the existing datasets, such as the objective, type of distortions, year, availability, number of samples, and availability of subjective scores.

Most of the existing datasets are designed for speech quality assessment and lack other domains that are dominant in UGM, such as music, background sounds, casual conversations,

¹The dataset along with the code is [Online]. Available: <https://github.com/dmumtaz/ugmq>

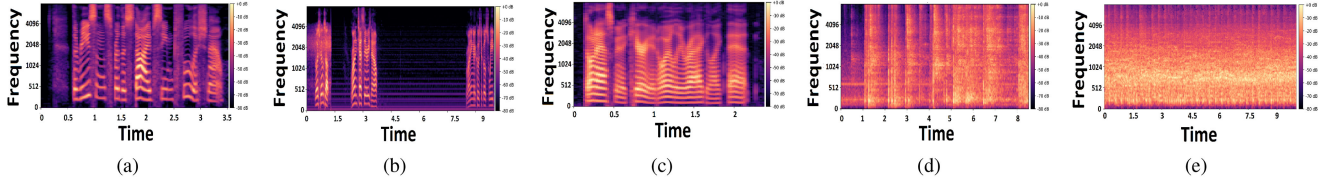


Fig. 1. Spectrogram of the samples from (a) NOIZUS [17], (b) SPINE [16], and (c) TIMIT [14]. (d) and (e) are two random instances from the IIT-JMU-UGM audio dataset.

interviews, etc. Prior work explored UGM audio quality estimation [24], wherein clean samples were acquired from YouTube, which were later corrupted by background noise. However, the repository is not openly available for use, and the number of samples is also low. In general, UGM audio signals have different acoustic characteristics as compared to plain speech content, and consequently, the quality assessment algorithm designed for speech signals cannot be directly employed on UGM. In order to quantitatively verify these arguments, we carried out spectrogram analysis of speech signals and UGM clips.

The spectrogram aids in analyzing the range of frequency and amplitude present in a given signal. It visualizes a signal on two perpendicular axes representing the time and the frequency, while the color intensity represents the amplitude. Given a signal $s[m]$ with window w , such that Ω is the frequency and m is the discrete-time index. The short-time Fourier transform (STFT) is expressed as

$$\begin{aligned} \text{STFT}\{s[m]\}(n, \Omega) &= S(n, \Omega) \\ &= \sum_{m=-\infty}^{\infty} s[m]w[m-n]e^{-j\Omega m}. \end{aligned} \quad (1)$$

Squaring the magnitude of the resultant STFT renders the spectrogram display of the power spectral density of the function as given by

$$\text{spectrogram}\{s(t)\}(\tau, \Omega) = |S(\tau, \Omega)|^2 \quad (2)$$

where τ is the time axis.

Fig. 1(a)–(c) represents the spectrogram of samples obtained from the three datasets NOIZUS [17], SPINE [16], and TIMIT [14] containing speech samples, respectively. It can be observed from the spectrograms that speech is represented by a restrained spectrum along with precisely defined predictable perceptual components. Additionally, the presence of phonemes is quite evident from the discontinuities (dark areas) present in the plots. Furthermore, it can be observed that the spectrograms depict a restricted range of frequencies, due to the presence of human voice, which encompasses a limited frequency span. The heat map depicting amplitude variation is quite constrained for the given samples, thus giving us the intuition that speech samples have limited diversity in frequency and amplitude.

Next, we analyze the existing datasets based on context diversity. As clear from Table I, most of the available datasets mainly consist of speech or music samples, while the datasets proposed in [22] and [24] consist of a combination of a few audio types, yet the amount of diversity in all these repositories is minimal.

Thus, these datasets may not represent content from real-world scenarios.

Based on the above, we identify the following limitations with the existing datasets: 1) lack of diversity in terms of magnitude and distortion types; 2) lack of subjectively annotated and openly available UGM datasets; and 3) lack of diversity in terms of content and context. Thus, there is a need for a dataset that quintessentially represents UGM. To fill this gap, we propose a new UGM dataset, namely, IIT-JMU-UGM audio dataset, along with corresponding subjective scores.

III. IIT-JMU-UGM AUDIO DATASET

The proposed IIT-JMU-UGM audio dataset contains 1150 audio clips with their corresponding subjective scores. It comprises different content, context, and levels of distortions. The following steps were undertaken during the development of this dataset.

A. Data Collection and Preprocessing

For the initial data collection, we made use of the various multimedia sharing platforms, such as Flickr, YouTube, CVDL [31], and Vimeo. These contain a rich pool of UGM data uploaded by amateur users as well as professionals. Initially, about 350 audio–video samples were obtained, out of which 217 clips were identified, which had appropriate copyright permission, adequate amount of audio, and diverse content. During preprocessing, only the audio part of the sample was extracted from the multimedia clips. The average length of samples is about 8 s. In order to generate an extensive repository of audio instances, which mimics realistic scenarios, we incorporated various deformities that perceptually affect the audio quality. Consequently, we modeled the following two types of impairments.

- 1) *Background noise*: Background noises, such as Gaussian noise, pink noise, and impairments like hum, glitch, babble, microphone, echo, etc., may be present in a UGM clip, which subsequently degrade their perceptual quality. Various types and intensities of these noises were convolved with the clean samples to get different levels of degradation.
- 2) *Bit rate*: According to Winkler and Faller [32], the audio bit rate highly affects the perceptual quality, i.e., the lower the bit rate, the poorer the quality. In this view, the audio clips were compressed to a wide range of bit rates from very low to high (approximately 10 to 280 kb/s).

Finally, from the selected reference multimedia samples, we developed the final IIT-JMU-UGM audio dataset of 1150 audio

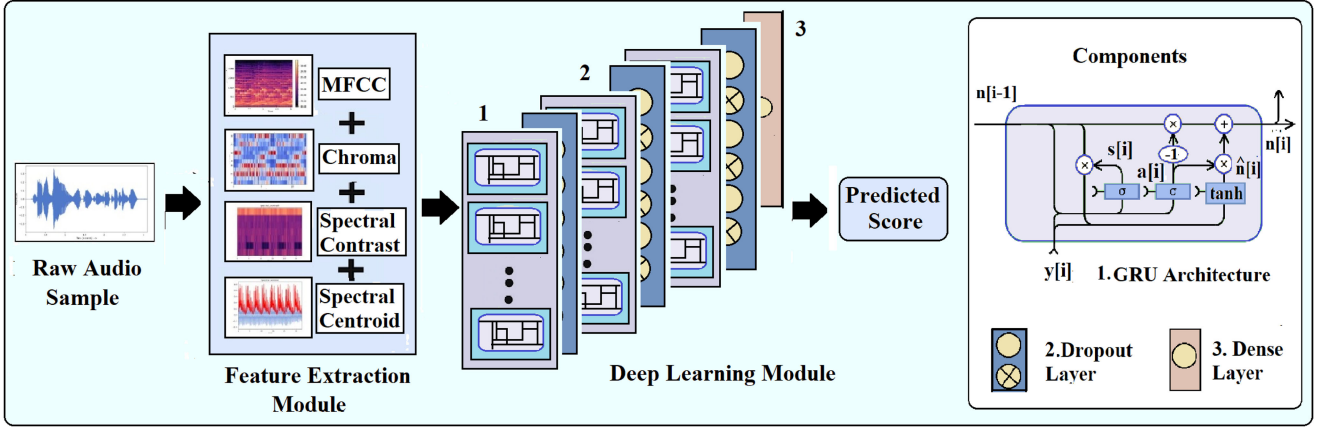


Fig. 2. Detailed architecture of the proposed algorithm.

clips, representing generalized and realistic UGM conditions to evaluate the subjective quality.

B. Testing Methodology

For subjective evaluation of the dataset, an online web application was designed. The single-stimulus model was adopted to record the subjective quality scores of the sequences. In this model, the subject is provided a single sample at a time without giving any information about the corresponding reference sample. Twenty-six volunteers participated in subjective testing. In the first step, basic demographic information about the subjects, such as age, gender, expertise in audio processing, the audio device used, etc., was collected. Furthermore, a brief training session was conducted, wherein a few audio samples were played to the subjects. It made the listeners acquainted with the user interface and the test procedure. The subjects had to rate the quality of the audio clips on a five-grade absolute category rating scale with “Bad,” “Poor,” “Fair,” “Good,” and “Excellent,” corresponding to 1 to 5 numeric score, respectively. Although no separate hearing test was conducted, yet having an average hearing ability was a prerequisite. It was also ensured that the audio devices did not produce any explicit sound degradation. In order to curtail the effects of fatigue, only 30 clips were made available to the listener at a given time, and then, they were asked to take a break before conducting further evaluations.

Eventually, in order to determine the final subjective rating of each audio clip, the mean opinion score (MOS) was computed by averaging the score given by the subjects for a particular clip, given by

$$MOS_i = \frac{1}{N_j} \sum_{j=1}^j X_{ij} \quad (3)$$

where X_{ij} represents the score given by the j th subject to the i th audio clip and N_j denotes the total number of ratings given to the clip. The result analysis (see Section V) discusses more descriptive statistics of the proposed IIT-JMU-UGM audio dataset. In the result analysis (see Section V), more details about

the proposed IIT-JMU-UGM audio dataset and the analytical study are discussed.

IV. PROPOSED ALGORITHM

This section proposes a non-intrusive quality assessment metric based on the deep learning technique for evaluating the perceptual quality of audio samples. The proposed method consists of two segments: the feature extraction module and the deep learning module for quality score prediction. Fig. 2 depicts the architecture of the proposed model along with the description of its components.

A. Feature Extraction Module

The first step in the proposed architecture is the extraction of key audio features. Feature extraction is the process of obtaining the most vital, discriminating, and compact representation of the raw data. These features reduce the input dimensionality while describing the original signal efficiently with respect to the given application. The reduced representation of the input data improves the time as well as the computational complexity of the learning algorithm, making it apt for real-time applications. We made use of Python’s Librosa library for feature extraction; the window length (fftsize) equal to 2048 was considered, the hop length was set to 512, and discrete cosine transform Type-2 along with the default sample rate was set for each clip. This article evaluates various feature extraction techniques, such as Mel-frequency cepstral coefficient (MFCC), spectral centroid, chroma, and spectral contrast. These features have been employed in several audio processing applications, such as speaker recognition, classification, information retrieval, speech enhancement, etc. Further details about these features can be found in [33].

- 1) **MFCC**: From the literature, it has been analyzed that MFCCs produce better results as compared to the various feature metrics for audio data since these are biologically inspired. The MFCCs depict the short-term power spectrum of sound instance, which is based upon a linear cosine transform of a log power spectrum on a nonlinear

Mel scale of frequency [34]. We made use of the first 20 MFCC features for the model.

- 2) *Spectral centroid (S)*: The spectral centroid represents the point where the energy of a spectrum is centered upon. It is extensively employed as an automatic estimator of timber in digital music and audio processing. It is obtained by calculating the weighted mean of the frequencies extracted using a Fourier transform, where the magnitudes are present as the weights.
- 3) *Chroma (C)*: The intensity value of the 12 defined musical chroma of the octave at each time frame is obtained by this feature [35]. In the feature extraction, the complete information regarding spectra corresponding to a given pitch class is accumulated into a unique coefficient.
- 4) *Spectral contrast (SC)*: The spectral contrast consists of the spectral valley, spectral peak, and their difference in every subband of the frequency. Seven SC features were used in this article.

For each audio sample, all the extracted features were concatenated into a single feature vector of size equal to 40. This vector is fed to the deep learning model for training.

B. Deep Learning Module

After estimating the relevant features, we explored the technique of deep learning to determine the perceptual quality of the IIT-JMU-UGM audio samples. Pioneering work in end-to-end time-series-based learning has been established using the RNN framework. The RNN is a type of artificial neural network, which makes use of time series or sequential data for predictions. RNNs can handle both static data like images and time-dependent series comprising sequential characteristics, such as video and audio. Thus, owing to the presence of recurrent connections, RNNs have proved to perform better than the regular DNNs and convolutional neural networks in time-series prediction [36]. However, one of the drawbacks of the basic RNN model is the vanishing gradient problem, which limits its performance. Thus, newer generations of RNNs have been developed to remove this problem. These include long short-term memory (LSTM), bidirectional long short-term memory (BiLSTM), and GRUs [37]. The LSTM comprises blocks containing cells. These cells consist of an input gate, an output gate, and a forget gate. The forget gate determines how certain values remain in the cell, thus adaptively resetting and forgetting the memory. The other two gates regulate the flow of data into and out of the cell, respectively. Alternatively, the BiLSTM contains two LSTMs: one sends the input in the forward direction, while the other transmits in the backward direction. Furthermore, the architecture of the GRU is similar to that of the LSTM; however, the GRU has less number of gates and fewer training parameters. This makes the GRU network faster and requires less memory [38].

Moreover, the stacked/deep RNN architecture consists of a series of RNN layers concatenated one after the other. The output from one layer serves as an input to the proceeding layer. Each layer fuses the learned representations of the earlier layer and feeds these to the next higher layer. Thus, better representations of the given data are obtained by the model [39]–[41].

TABLE II
MODEL SUMMARY OF THE PROPOSED ARCHITECTURE

Layer	Hidden Units	Dropout Rate	Output Shape	Parameters
GRU-1	64	—	(None, 128, 64)	20160
Dropout	—	0.2	(None, 128, 64)	—
GRU-2	32	—	(None, 128, 32)	9312
Dropout	—	0.2	(None, 128, 32)	—
GRU-3	8	—	(None, 8)	984
Dropout	—	0.2	(None, 8)	—
Dense	1	—	(None, 1)	9

In the literature, GRUs have been used in various speech signal modeling, natural language processing tasks, and music modeling [42], [43]. Based on the aforementioned consideration, in this article, we tailor a deep learning model using a stacked GRU architecture for audio quality assessment. The following equations define the standard GRU architecture [37]:

$$a_i = \sigma_k(W_a y_i + V_a n_{i-1} + b_a) \quad (4)$$

$$s_i = \sigma_k(W_r y_i + V_r n_{i-1} + b_s) \quad (5)$$

$$\hat{n}_i = \phi_n(W_n y_i + V_n (s_i \odot n_{i-1}) + b_n) \quad (6)$$

$$n_i = (1 - a_i) \odot n_{i-1} + a_i \odot \hat{n}_i \quad (7)$$

where a_i and s_i correspond to the update and the reset gate vectors, respectively. Furthermore, y_i and n_i represent the input and output vectors, respectively. The candidate activation vector is given by $\hat{n}_i$. The parameters for the feedforward connections are given as W_a , W_s , and W_n , while the parameters corresponding to the recurrent weights are given as V_a , V_s , and V_n . It consists of trainable bias vectors b_a , b_s , and b_n , which are added before applying the nonlinearities. The elementwise multiplication is represented by $\odot$. Activation functions of both the update and reset gates are the sigmoidal function σ_k such that $\sigma_k(y) \in [0, 1]$, while ϕ_n is the hyperbolic tangent function. The time complexity for one gradient step for the sequence length M is given by $\mathcal{O}(M \times q_p + M \times q_p \times q_r)$, where q_p and q_r are the hidden state and the input dimensions [44], respectively.

The feature vector obtained in the first step is fed to the designed stacked GRU model. We made use of three GRU layers (GRU-1, GRU-2, and GRU-3) stacked on top of one other, each followed by a Dropout layer. The Dropout layer is mainly used to prevent overfitting as it randomly sets input units to zero at each step during the training. The number of hidden units in each of the GRU layers is set to 64, 32, and 8, respectively. Finally, a Dense layer with one output linear function is added to obtain the final quality scores. The output value corresponds to the quality score from the range 1–5 (bad to excellent). The summary of the proposed model is given in Table II.

To determine the performance of different optimizers, such as Adam, root-mean-square propagation (RMSProp), AdaDelta, and stochastic gradient descent (SGD), on the proposed model, we conducted an ablation study. As clear from Table III, the Adam optimizer works best for the model followed by RMSProp. The Adam optimizer is a combination of the gradient descent with the momentum algorithm and the RMSProp algorithm. It has the characteristics of being computationally

TABLE III

PERFORMANCE COMPARISON OF THE PROPOSED STACKED GRU ARCHITECTURE USING DIFFERENT OPTIMIZATION ALGORITHMS IN TERMS OF PLCC, SRCC, KRCC, AND RMSE

Optimizer	PLCC	SRCC	KRCC	RMSE
ADAM	0.834	0.818	0.625	0.776
RMS Prop	0.821	0.798	0.610	0.779
AdaDelta	0.818	0.795	0.611	0.799
SGD	0.798	0.786	0.603	0.802

TABLE IV

PERFORMANCE COMPARISON OF THE PROPOSED STACKED GRU ARCHITECTURE USING DIFFERENT LOSS FUNCTIONS IN TERMS OF PLCC, SRCC, KRCC, AND RMSE

Loss Function	PLCC	SRCC	KRCC	RMSE
MSE	0.834	0.818	0.625	0.776
MSLE	0.815	0.785	0.601	0.776
MAE	0.796	0.768	0.588	0.858

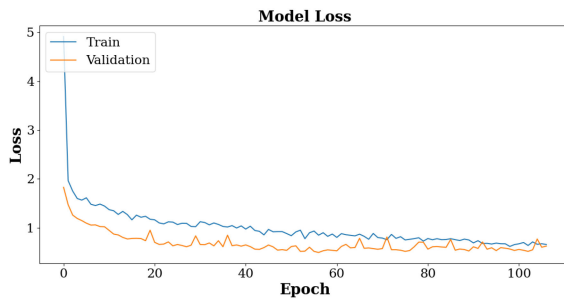


Fig. 3. Plot between the loss and the number of epochs.

efficient, converging easily, simple to implement, and requiring less memory, thus giving good results for the task.

Furthermore, the performance of different loss functions, such as mean square error (MSE), mean absolute error (MAE), and mean squared logarithmic error (MSLE), is also analyzed. The results in Table IV show that the MSE works better as compared to the other techniques. The MSE is a simple yet efficient function, which calculates the square of the difference between the actual and the predicted values.

One of the most vital issues that should be taken care of in deep learning is overfitting. In this regard, we made sure that the proposed model, along with the hyperparameters, gives optimal results without overfitting. Figs. 3 and 4 showcase the plots between loss and number of epochs, and PLCC and number of epochs, respectively. As the model trains over the epoch, the loss decreases, and the performance (PLCC) increases. Initially, there is a drastic decrease in the loss value; however, after some iterations, as the loss does not decrease any further, it is saved as the final model. The model converges at approximately 100 epochs. We used early stopping to stop the model from further training if the loss does not decrease after 50 epochs. It is clearly evident from these plots that the model is converging gradually without overfitting.

In the final model, the MSE function was employed to minimize the loss between the network outputs and the training

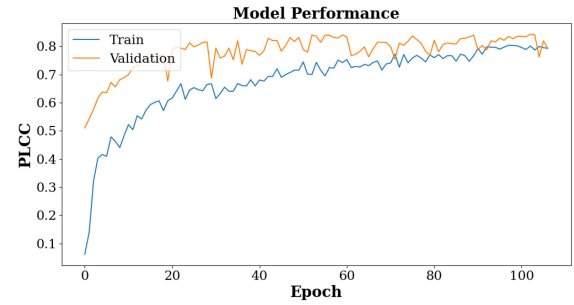


Fig. 4. Plot between the PLCC and the number of epochs.

labels. The dropout value was set to 0.2, while the number of epochs was equal to 300. Furthermore, we experimented with different batch sizes (8, 16, 32, 64, and 128) and obtained the optimal performance for the batch size equal to 16. We used the Adam optimizer with the learning rate equal to 0.01, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and the epsilon factor set to $1e-8$. Furthermore, 70% of IIT-JMU-UGM audio samples were selected for training, 15% for testing, and the remaining 15% for validation purposes. We also employed other types of RNN, whose performance is described in the next section.

The proposed metric was implemented in Python 3 and mainly used NumPy and Librosa libraries apart from Keras with TensorFlow background. Moreover, the model was trained and tested on DELL G3 15 i7 core, eighth generation, and Nvidia GTX GPU. In this environment, it takes about 18 min to extract the audio features and train the model. The average time for predicting the quality of a single clip of 8-s duration is about 0.520 s.

V. RESULT ANALYSIS

In this article, we first designed and developed the novel IIT-JMU-UGM audio dataset, and second, we developed a deep-learning-based model for the quality assessment of UGM audio data. In this view, we evaluate both these aspects in the following subsections.

A. Analysis of the Proposed IIT-JMU-UGM Audio Dataset

In order to show that the proposed IIT-JMU-UGM audio dataset is comprehensive and can overcome the issues associated with the existing datasets, we conducted various empirical studies.

Similar to the study conducted in Section II, we plot the spectrogram of two random samples of the IIT-JMU-UGM audio dataset in Fig. 1(d) and (e). Comparing these samples with the spectrograms of audio in the existing datasets [see Fig. 1(a)–(c)], it is evident that the samples in the proposed dataset are spread over a wide range of frequency spectrum. These samples comprise a mixture of music, background sounds, speech, and beats that have a broad spectral width.

Furthermore, to evaluate the diversity in the dataset, a histogram was used. Fig. 5 depicts the division of the dataset into 13 diverse context categories, such as tutorials, home videos,

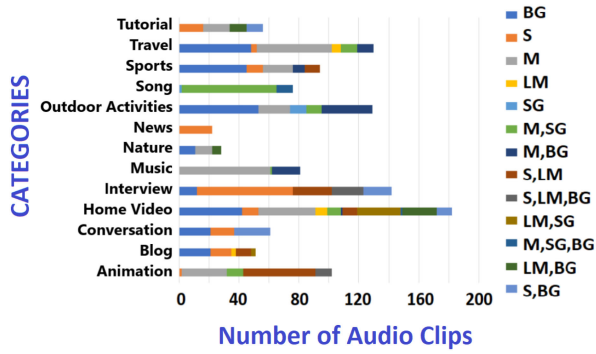


Fig. 5. Histogram representing various categories of context and content in the IIT-JMU-UGM audio dataset. Content is annotated as: song (SG), music (M), light music (LM), speech (S), and background sounds (BG).

songs, music, sports, outdoor activities, etc. These samples were acquired from numerous devices, such as smartphones, handheld devices, HD cameras, etc., by amateurs and professionals. These clips contain content, such as music, song, human voice, backdrop music, machine sound, animal sound, and other background sounds, represented in the plot by various colors in Fig. 5. Furthermore, apart from the explicitly added distortions (background noise and different bit rates), about 10% of these clips implicitly consist of several distortions caused by the varying quality of the capturing device, environmental factors, compression degradation, etc. With this view, the proposed dataset is significantly generic and showcases different real-world scenarios.

Next, in order to validate the consistency of different subjects during the subjective testing, we made use of the Fleiss' Kappa technique. This technique is a statistical measure to quantify the level of agreement between the different subjects. The result of this test came to be 0.405, which is interpreted as “Moderate agreement among the subjects,” thus validating the reliability of the subjective testing.

Additionally, as a part of the subjective testing, the subjects were also asked to choose the perceptually annoying degradation they notice in the audio clip: background noise, distorted sound (broken), or clear (no degradation). The histogram plot (see Fig. 6) between the MOS and the type of perceptual annoyance indicates that low-MOS clips also had the maximum number of perceptual distortions and *vice versa*. This further strengthened that the subjects were highly involved in the testing, and thus, the subjective testing is reliable.

Fig. 7(a) represents the MOS distribution over the entire dataset. As observed from the figure, the plot exhibits a diverse range of quality scores. Fig. 7(b) showcases the histogram of the MOS categorized into five discrete quality levels (1–5). As prominent from the plot, the dataset represents the various quality levels in a fairly balanced ratio. These empirical studies validate that the proposed dataset is comprehensive and generalized.

Fig. 8 represents a scatter plot between the audio bit rate and the MOS. As per the general perception, the audio quality is

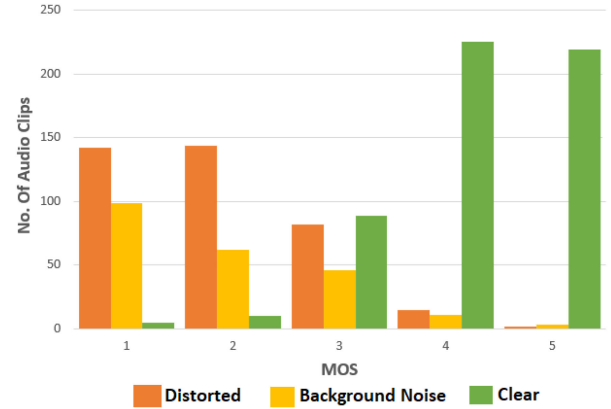


Fig. 6. Histogram of the MOS and the type of perceptual annoyance.

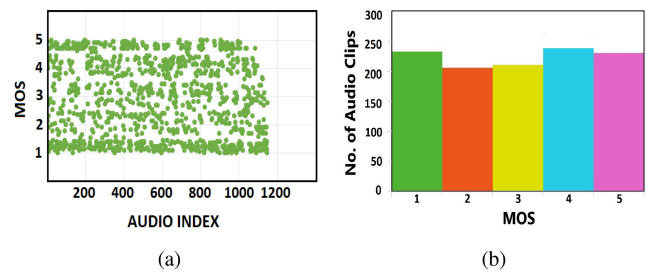


Fig. 7. (a) Overall distribution of the MOS. (b) MOS distribution into five discrete quality classes on the IIT-JMU-UGM audio dataset.

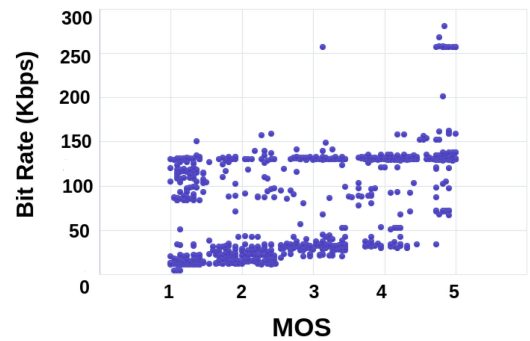


Fig. 8. Scatter plot between the bit rate and the MOS.

directly proportional to the bit rate, i.e., the lower the bit rate, the poorer the audio quality and *vice versa*. However, from the plot, it is observed against the notion. Other parameters, such as background sound, context, noise, etc., may also significantly impact the perceptual quality. This observation further emphasizes on developing a generalized quality metric for evaluating UGM audio data.

B. Result Analysis of the Proposed Stacked GRU Model

In this section, we analyze the performance of the proposed method in two ways. First, we perform the ablation study of various hyperparameters used in the model. Next, we compare the proposed model with the existing state-of-the-art techniques.

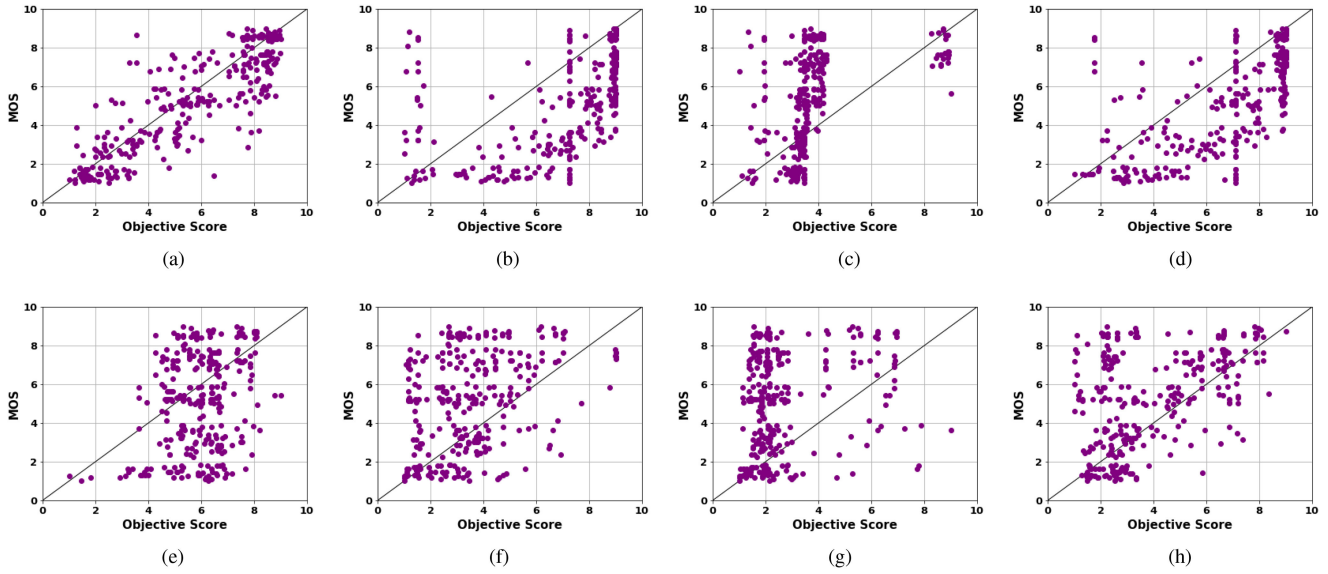


Fig. 9. Scatter plot between the MOS and objective scores of different quality assessment metrics. (a) Proposed model. (b) STOI [7]. (c) SISDR [8]. (d) PESQ [4]. (e) SNRVAD [13]. (f) SRMR [11]. (g) NIST-STNR [12]. (h) NISQA [9].

TABLE V

PERFORMANCE COMPARISON OF THE PROPOSED STACKED GRU ARCHITECTURE USING DIFFERENT NUMBER OF GRU LAYERS IN TERMS OF PLCC, SRCC, KRCC, AND RMSE

Number of Layers	PLCC	SRCC	KRCC	RMSE
1	0.605	0.603	0.426	1.052
2	0.736	0.703	0.518	0.971
3	0.834	0.818	0.625	0.776

TABLE VI

PERFORMANCE COMPARISON OF THE PROPOSED STACKED GRU ARCHITECTURE WITH DIFFERENT RNN MODELS IN TERMS OF PLCC, SRCC, KRCC, AND RMSE

Model	PLCC	SRCC	KRCC	RMSE
GRU	0.834	0.818	0.625	0.776
LSTM	0.801	0.778	0.595	0.816
BiLSTM	0.761	0.736	0.545	0.860
Simple RNN	0.617	0.554	0.423	1.073

For this, we employ the standard correlation evaluation methods, which include PLCC, Kendall's rank correlation coefficient (KRCC), Spearman's rank order correlation coefficient (SRCC), and root-mean-square error (RMSE). PLCC is a statistical measure for predicting linearity, SRCC describes the relationship utilizing a monotonic function, KRCC calculates the amount of dependence between two variables, and the RMSE is used to determine the error of a model in predicting the data. A metric is said to perform well if the value of PLCC, SRCC, and KRCC is close to 1, while the RMSE should tend to 0. The scores obtained from various quality assessment algorithms are mapped to the ground truth or the MOS using the following equation of nonlinearity [47]:

$$a(B) = c_1 \left(\frac{1}{2} - \frac{1}{1 + e^{c_2(B-c_3)}} \right) + c_4 B + c_5 \quad (8)$$

where B and $a(B)$ are the objective and its correspondingly mapped subjective score, respectively, while c_1, c_2, c_3, c_4 , and c_5 refer to the parameters required to be fitted during nonlinear regression.

In the proposed model, we made use of the stacked GRU architecture, which is three-layer deep. The rationale behind using the particular number of levels in the model is depicted in Table V. The ablation study consists of the performance of the model using different number of GRU layers. As depicted from

the results, by using one or two layers of the GRU, the model is able to achieve 0.605 and 0.736 PLCC values. If we increase the number of GRU levels to three, there is an improvement in the PLCC scores to 0.834. As depicted from Table V, it was observed that the deep GRU model gives better results than the shallow ones in terms of PLCC, SRCC, KRCC, and RMSE.

We compare our proposed stacked GRU model with various available RNN types, such as simple RNN, LSTM, and BiLSTM, in order to determine the relative performance. For the basic RNN model, we used three simple RNN layers with 64, 32, and 8 hidden units, with the dropout equal to 0.4, 0.3, and 0.3, respectively. For the stacked LSTM model, we combined three LSTM layers of size 64, 32, and 8, the dropout was set to 0.05, and the recurrent dropout was set to 0.35 for all the layers. For the stacked BiLSTM model, we used two layers, with the dropout equal to 0.4 and 0.35 and the recurrent dropout equal to 0.35 for both the layers. The final quality score was acquired from the last Dense layer of size equal to 1 in all the models. All the models were k -fold validated, and the standard measures to prevent overfitting were employed. As depicted from Table VI, the stacked GRU performs better than the other RNN models. In addition, it was observed that the proposed GRU model converged in about 18% less time as compared to the LSTM and about 23% less time as compared to the BiLSTM.

TABLE VII
PERFORMANCE COMPARISON OF THE PROPOSED ALGORITHM AGAINST VARIOUS AUDIO QUALITY METRICS IN TERMS OF PLCC, SRCC, KRCC, RMSE, AND F-SCORE

S. No.	Metric	Type	PLCC	SRCC	KRCC	RMSE	F-Score
1.	Proposed Metric-GRU	Non-Intrusive	0.834	0.818	0.625	0.776	-
2.	Wawenets [45]	Non-Intrusive	0.683	0.633	0.458	0.815	+1
3.	SRMR [11]	Non-Intrusive	0.297	0.293	0.197	1.192	+1
4.	NIST-STNR [12]	Non-Intrusive	0.273	0.255	0.174	1.276	+1
5.	SNRVAD [13]	Non-Intrusive	0.228	0.208	0.140	1.274	+1
6.	MOSNET [26]	Non-Intrusive	0.210	0.200	0.131	1.278	+1
7.	WADA [46]	Non-Intrusive	0.119	0.144	0.098	1.336	+1
1.	ViSQOL [10]	Intrusive	0.734	0.715	0.523	0.736	+1
2.	STOI [7]	Intrusive	0.704	0.652	0.480	0.924	+1
3.	SISDR [8]	Intrusive	0.697	0.652	0.475	1.020	+1
4.	PESQ [4]	Intrusive	0.683	0.641	0.468	1.207	+1
5.	NISQA [9]	Intrusive	0.410	0.332	0.232	1.106	+1

The table is arranged in descending order of PLCC.

Thus, the GRU outperformed the other RNNs in terms of not only PLCC but also computation time.

Next, we evaluated the performance of the proposed method with various available state-of-the-art quality assessments methods. For this purpose, five intrusive metrics, i.e., SISDR [8], NISQA [9], ViSQOL [10], PESQ [4], and STOI [7], were considered. In addition, six non-intrusive metrics, i.e., SRMR [11], Wawenets [45], MOSNET [26], NIST-STNR [12], WADA [46], and SNRVAD [13], were considered. Table VII illustrates the performance of various objective quality metrics on the IIT-JMU-UGM audio dataset. The results show that the proposed deep learning method yields better performance than the existing algorithms, with the PLCC equal to 0.834, SRCC equal to 0.810, and KRCC equal to 0.683. It outperforms all the intrusive metrics, i.e., ViSQOL, STOI, SISDR, PESQ, and NISQA. One of the drawbacks of the highest performing metric ViSQOL as per its description is that it is designed for specific sample rates (48 kHz). Thus, it is not suitable for all types of audio, which limits its usage. On the other hand, the PESQ metric is computationally demanding. The performance of the proposed method also surpasses all the mentioned non-intrusive approaches, i.e., Wawenets, SRMR, NIST-STNR, SNRVAD, MOSNET, and WADA. Again, Wawenets is able to evaluate the clips with sample rate equal to 45 kHz only, limiting its usability. Overall, our proposed model beats the benchmark metrics. This is because these methods were mainly devised for speech quality assessment, while the proposed IIT-JMU-UGM audio dataset is quite generalized and consists of diverse sounds present in the user-generated data.

Additionally, in order to verify the statistical significance of the proposed metric, we conducted F -test between the proposed metric and the existing methods. The output “+1” depicts that the proposed method is better, “−1” signifies that it is worse, and “0” represents it is equally comparable to the other metric [47]. The last column of Table VII represents the F -score between the state-of-the-art metric and the proposed metric. It can be observed that the F -score is “+1” in all cases. This makes it

evident that the proposed audio quality method is statistically better than all the other state-of-the-art metrics.

Finally, Fig. 9 illustrates the scatter plot of the objective score and the corresponding subjective scores (MOS) of the samples obtained by some of the quality assessment metrics, which include STOI, SISDR, PESQ, SNRVAD, SRMR, NIST-STMR, and NISQA. For better visualization, randomly, 300 samples were plotted. It can be intuitively observed that the scatter plot of the proposed model depicts high linearity and converges more efficiently than the other metrics. This indicates that the proposed model scores map much better with the subjective score.

From all the above experiments, it can be validated that the existing quality assessment metrics are not able to correlate well with the subjective evaluations for various types of distortions in UGM audio samples. On the other hand, the proposed quality assessment metric performs better than the existing algorithms and efficiently mirrors human auditory perception.

VI. CONCLUSION

To facilitate quality assessment of UGM audio, we first developed a novel IIT-JMU-UGM audio dataset comprising diverse content, context, and degrees of distortions typically present in various real-time multimedia applications. Then, human subjective testing was conducted on the dataset to obtain ground truth on quality. On careful investigation, it was deduced that though the degradations caused by low bit rate have a significant effect on the perceived audio quality, there may not be a linear relationship between the two. Other factors, such as context, content and background sounds, etc., also have a profound effect on the overall quality. To encapsulate all the factors, we proposed a robust non-intrusive stacked GRU architecture for estimating the quality of audio. We inputted perceptually important audio features, such as MFCC, spectral centroid, chroma, and spectral contrast, as input to the model. The proposed non-intrusive quality metric outperformed the existing state-of-the-art

intrusive and non-intrusive methods applied on the IIT-JMU-UGM audio dataset. In the future, integration of advanced learning techniques (such as transfer learning), the influence of context, and the combined effect of other media modes (audio, video, and text) on the overall perceived multimedia quality can be investigated.

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